Lecture 10: Section 3.1

Implicit Functions

Md Ahsanul Haque, Ph.D.

Assistant Professor
Department of Mathematics and Physics
North South University, Dhaka

Explicit vs Implicit Functions

Explicit Functions

y is expressed **directly** in terms of x:

- $y = x^2 + 3x - 1$
- $y = \sqrt{x+2}$
- $y = \dfrac{1}{x^2+1}$

Form: $y = f(x)$

Implicit Functions

Relationship between x and y is given **indirectly**:

- $x^2 + y^2 = 25$
- $x^3 + y^3 = 6xy$
- $\sin(x+y) = y^2 \cos x$

Form: $F(x, y) = 0$

Example

- **Explicit:** $y = \sqrt{25 - x^2}$ (top half of circle)
- **Implicit:** $x^2 + y^2 = 25$ (entire circle)

Why Implicit Differentiation?

The Problem

Some equations are difficult or impossible to solve explicitly for y:

- $x^3 + y^3 = 6xy$
- $y^4 + 2y^2x - 3yx^2 + x^3 = 0$
- $\sin(xy) = x + y$

But we still want to find $\frac{dy}{dx}$!

The Solution: Implicit Differentiation

Differentiate both sides with respect to x, treating y as a function of x, then solve for $\frac{dy}{dx}$.

Key Insight

Even if we can't write $y = f(x)$ explicitly, we can still find how y changes with respect to x!

The Basic Idea: Chain Rule in Disguise

Theorem (Implicit Differentiation)

If y is a function of x defined implicitly by $F(x, y) = 0$, then:

$$\frac{d}{dx}[y] = \frac{dy}{dx} \quad \textit{but} \quad \frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$$

For y^2:

- Think: $y^2 = [y(x)]^2$
- By chain rule: $\frac{d}{dx}[y^2] = 2y\frac{dy}{dx}$

The Process

1. Differentiate both sides with respect to x
2. Use chain rule for y-terms: $\frac{d}{dx}[f(y)] = f'(y)\frac{dy}{dx}$
3. Collect $\frac{dy}{dx}$ terms
4. Solve for $\frac{dy}{dx}$

Basic Implicit Differentiation Examples

Example

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$

Solution

$$\frac{d}{dx}[x^2 + y^2] = \frac{d}{dx}[25]$$

$$2x + 2y\frac{dy}{dx} = 0$$

$$2y\frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Basic Implicit Differentiation Examples

Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$

$$\frac{d}{dx}[x^3 + y^3] = \frac{d}{dx}[6xy]$$

$$3x^2 + 3y^2\frac{dy}{dx} = 6\left(y + x\frac{dy}{dx}\right)$$

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$$

$$3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 3x^2$$

$$(3y^2 - 6x)\frac{dy}{dx} = 6y - 3x^2$$

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

The derivative $\frac{dy}{dx}$ is often expressed in terms of **both** x and y.

Practice 1: Basic Implicit Differentiation

Find $\frac{dy}{dx}$ for each equation:

1. $x^2 + y^2 = 36$
2. $x^3 - y^3 = 2xy$
3. $y^4 + x^2y^2 = 10$
4. $\sqrt{x} + \sqrt{y} = 5$

Strategy

1. Differentiate both sides with respect to x
2. Remember: $\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$
3. For products: $\frac{d}{dx}[xy] = y + x\frac{dy}{dx}$
4. Solve for $\frac{dy}{dx}$

Practice 1 Solutions

1. $x^2 + y^2 = 36$
   $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$
2. $x^3 - y^3 = 2xy$
   $3x^2 - 3y^2\frac{dy}{dx} = 2y + 2x\frac{dy}{dx}$
   $-3y^2\frac{dy}{dx} - 2x\frac{dy}{dx} = 2y - 3x^2$
   $\frac{dy}{dx} = \frac{3x^2-2y}{3y^2+2x}$
3. $y^4 + x^2y^2 = 10$
   $4y^3\frac{dy}{dx} + 2xy^2 + x^2 \cdot 2y\frac{dy}{dx} = 0$
   $(4y^3 + 2x^2y)\frac{dy}{dx} = -2xy^2$
   $\frac{dy}{dx} = -\frac{2xy^2}{4y^3+2x^2y} = -\frac{xy}{2y^2+x^2}$
4. $\sqrt{x} + \sqrt{y} = 5$
   $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$
   $\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$

Differentiability of Implicit Functions

Theorem (Implicit Function Theorem - Simplified)

If $F(x, y) = 0$ defines y implicitly as a function of x near a point (a, b), and $\frac{\partial F}{\partial y} \neq 0$ at (a, b), then y is differentiable at $x = a$.

For most "nice" equations, implicit differentiation works. But be careful at points where:

- The denominator in $\frac{dy}{dx}$ is zero
- Vertical tangents occur

Example

For $x^2 + y^2 = 25$, we found $\frac{dy}{dx} = -\frac{x}{y}$

- At $y = 0$ (points $(5, 0)$ and $(-5, 0)$), derivative is undefined
- These are points with vertical tangents

Finding Tangent Lines Implicitly

Find the equation of the tangent line to $x^2 + xy + y^2 = 7$ at the point $(1, 2)$

1. Differentiate implicitly:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$$

$$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

2. Solve for $\frac{dy}{dx}$:

$$(x + 2y)\frac{dy}{dx} = -2x - y$$

$$\frac{dy}{dx} = \frac{-2x - y}{x + 2y}$$

3. At $(1, 2)$: $\frac{dy}{dx} = \frac{-2(1)-2}{1+2(2)} = \frac{-4}{5}$
4. Tangent line: $y - 2 = -\frac{4}{5}(x - 1)$

Practice 2: Tangent Lines and Specific Points

1. Find $\frac{dy}{dx}$ for $x^2y + xy^2 = 6$ and evaluate at $(1, 2)$
2. Find the tangent line to $y^2 = x^3 - 3x + 2$ at $(2, 2)$
3. For $\sin(xy) = x$, find $\frac{dy}{dx}$ and evaluate at $(\frac{\pi}{2}, 1)$
4. Find where the tangent line is horizontal for $x^2 + 4y^2 = 16$

Tips

- Substitute point coordinates **after** differentiating
- Horizontal tangent: set numerator of $\frac{dy}{dx} = 0$
- Vertical tangent: set denominator of $\frac{dy}{dx} = 0$

Practice 2 Solutions

1. $x^2y + xy^2 = 6$
   $2xy + x^2\frac{dy}{dx} + y^2 + 2xy\frac{dy}{dx} = 0$
   At $(1,2)$: $4 + \frac{dy}{dx} + 4 + 4\frac{dy}{dx} = 0$
   $5\frac{dy}{dx} = -8 \Rightarrow \frac{dy}{dx} = -\frac{8}{5}$
2. $y^2 = x^3 - 3x + 2$
   $2y\frac{dy}{dx} = 3x^2 - 3$
   At $(2,2)$: $4\frac{dy}{dx} = 9 \Rightarrow \frac{dy}{dx} = \frac{9}{4}$
   Tangent: $y - 2 = \frac{9}{4}(x-2)$
3. $\sin(xy) = x$
   $\cos(xy)\left(y + x\frac{dy}{dx}\right) = 1$
   At $(\frac{\pi}{2}, 1)$: $\cos(\frac{\pi}{2})\left(1 + \frac{\pi}{2}\frac{dy}{dx}\right) = 1$
   $0 = 1 \rightarrow$ No tangent line (point not on curve!)
4. $x^2 + 4y^2 = 16 \Rightarrow \frac{dy}{dx} = -\frac{x}{4y}$
   Horizontal when numerator $= 0$: $x = 0 \Rightarrow y = \pm 2$
   Points: $(0,2)$ and $(0,-2)$

Advanced Examples: Trigonometric and Exponential

Find $\frac{dy}{dx}$ for $\sin(x+y) = y^2 \cos x$

$$\frac{d}{dx}[\sin(x+y)] = \frac{d}{dx}[y^2 \cos x]$$

$$\cos(x+y)\left(1+\frac{dy}{dx}\right) = 2y\frac{dy}{dx}\cos x + y^2(-\sin x)$$

$$\cos(x+y) + \cos(x+y)\frac{dy}{dx} = 2y\cos x\frac{dy}{dx} - y^2 \sin x$$

$$\cos(x+y)\frac{dy}{dx} - 2y\cos x\frac{dy}{dx} = -y^2 \sin x - \cos(x+y)$$

$$\frac{dy}{dx}[\cos(x+y) - 2y\cos x] = -[y^2 \sin x + \cos(x+y)]$$

$$\frac{dy}{dx} = \frac{y^2 \sin x + \cos(x+y)}{2y\cos x - \cos(x+y)}$$

Second Derivatives Implicitly

Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 25$

We already have: $\frac{dy}{dx} = -\frac{x}{y}$

Differentiate both sides with respect to x:

$$\frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d}{dx}\left[-\frac{x}{y}\right]$$

$$\frac{d^2y}{dx^2} = -\frac{y - x\frac{dy}{dx}}{y^2}$$

Substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y - x(-\frac{x}{y})}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2}$$

$$= -\frac{y^2 + x^2}{y^3} = -\frac{25}{y^3}$$

Practice 3: Advanced Problems

1. Find $\frac{dy}{dx}$ for $e^{xy} = x + y$
2. Find $\frac{d^2y}{dx^2}$ for $x^2 - y^2 = 16$
3. Find $\frac{dy}{dx}$ for $\ln(x^2 + y^2) = 2\tan^{-1}(\frac{y}{x})$
4. Find where the tangent line is vertical for $y^2 = x^3 - 3x^2$

Advanced Tips

- For exponentials: $\frac{d}{dx}[e^u] = e^u \frac{du}{dx}$
- For second derivatives: differentiate the first derivative
- Vertical tangents occur when denominator of $\frac{dy}{dx} = 0$

Practice 3 Solutions

1. $e^{xy} = x + y$

   $e^{xy}\left(y + x\frac{dy}{dx}\right) = 1 + \frac{dy}{dx}$

   $\frac{dy}{dx} = \frac{1-ye^{xy}}{xe^{xy}-1}$

2. $x^2 - y^2 = 16$

   First derivative: $2x - 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y}$

   Second derivative: $\frac{d^2y}{dx^2} = \frac{y - x\frac{dy}{dx}}{y^2} = \frac{y - x(\frac{x}{y})}{y^2} = \frac{y^2 - x^2}{y^3} = -\frac{16}{y^3}$

3. $\ln(x^2 + y^2) = 2\tan^{-1}(\frac{y}{x})$

   $\frac{2x + 2y\frac{dy}{dx}}{x^2+y^2} = 2 \cdot \frac{1}{1+(\frac{y}{x})^2} \cdot \frac{x\frac{dy}{dx} - y}{x^2}$

   This simplifies to $\frac{dy}{dx} = \frac{x+y}{x-y}$

4. $y^2 = x^3 - 3x^2$

   $\frac{dy}{dx} = \frac{3x^2-6x}{2y}$

   Vertical when $y = 0$: $0 = x^3 - 3x^2 \Rightarrow x^2(x-3) = 0$

   Points: $(0, 0)$ and $(3, 0)$

Applications of Implicit Differentiation

Related Rates

- Changing shapes (circles, spheres, rectangles)
- Sliding ladders
- Filling/empting containers

Economics

- Production functions
- Indifference curves
- Isoquant curves

Physics

- Orbital mechanics
- Wave equations
- Thermodynamic relationships

Common Mistakes and Pitfalls

Forgetting the Chain Rule

- Wrong: $\frac{d}{dx}[y^2] = 2y$
- Correct: $\frac{d}{dx}[y^2] = 2y\frac{dy}{dx}$

Product Rule Errors

- Wrong: $\frac{d}{dx}[xy] = \frac{dy}{dx}$
- Correct: $\frac{d}{dx}[xy] = y + x\frac{dy}{dx}$

Algebraic Mistakes

- Not collecting all $\frac{dy}{dx}$ terms
- Forgetting to simplify final answer

Summary: Implicit Differentiation

When to Use It

- Equation relates x and y but can't solve for y explicitly
- Need derivative at specific point
- Working with curves defined implicitly

Key Steps

1. Differentiate both sides with respect to x
2. Use chain rule: $\frac{d}{dx}[f(y)] = f'(y)\frac{dy}{dx}$
3. Use product rule when needed
4. Collect all $\frac{dy}{dx}$ terms
5. Solve for $\frac{dy}{dx}$

Final Practice Problems

Find $\frac{dy}{dx}$ for each equation:

1. $x^3 + y^3 = 3xy$
2. $y \sin x = x \cos y$
3. $e^{x/y} = x - y$
4. $\sqrt{xy} = x^2 y + 1$
5. $\cos(x - y) = y \sin x$

Challenge Problems

6. Find $\frac{d^2y}{dx^2}$ for $xy + y^2 = 1$
7. Find the points where the tangent line is horizontal for $x^2 - xy + y^2 = 3$

Remember

Practice is key to mastering implicit differentiation!